

Mantavya Maan

B.S. Mathematics | Indian Institute of Technology Bombay
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Education

Indian Institute of Technology Bombay <i>B.S. in Mathematics; Minor in Game Theory and Logic, Department of Computer Science and Engineering</i>	CPI: 8.86/10 <i>Expected May 2027</i>
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Scholastic Achievements

- Secured top **0.49** percentile in **JEE Main 2023** examination out of **1.1 million** eligible candidates all over India.
- Among the top **0.97** percentile in **JEE Advanced 2023** examination out of **0.18 million** candidates all over India.

Professional Experience

Palpx <i>Summer Intern, Mathematical Modeling & Artificial Intelligence</i>	May 2026 – Jul 2026
- Formulated 4 predictive scoring models via statistical inference, probabilistic modeling, and mathematical optimization. - Applied lambda calculus, computational logic, functional modeling, and Retrieval-Augmented Generation to AI systems. - Architected an LLM orchestration platform supporting 100+ AI models using Bayesian inference, utility scoring, and routing.	
Gradpipe <i>Machine Learning & Data Engineering Intern</i>	Nov 2025 – Dec 2025
- Architected a high-dimensional talent discovery engine using vector databases and hybrid retrieval for semantic matching. - Engineered an AST-based analysis pipeline evaluating complexity, concurrency, and memory utilization for repository scoring. - Designed an LLM-as-a-Judge evaluation framework using standardized benchmarks, reproducible scoring, and model validation.	
Fino Payments Bank <i>Summer Intern, Data Analytics</i>	Jun 2025 – Jul 2025
- Analyzed 1M+ financial records using SQL, Pandas, and NumPy to identify market trends and evolving customer behavior. - Developed interactive Tableau dashboards tracking 20+ KPIs to accelerate executive decision-making and business reporting. - Collaborated across cross-functional teams to strengthen data integrity, reporting pipelines, and business analytics processes.	

Projects

Atlas Neural Gateway <i>AI Infrastructure & Multi-LLM Orchestration</i>	May 2026 – Jul 2026
- Architected an AI orchestration platform routing 100+ LLMs using Bayesian inference, Thompson sampling, and planning. - Designed a KNN vector-embedding parser using FastEmbed, ONNX, semantic retrieval, caching, and sub-10 ms inference. - Engineered a FastAPI control plane with OpenRouter synchronization, dynamic model registry, and adaptive policy enforcement. - Developed multi-stage execution planning supporting verifier selection, fallback routing, response streaming, and audit trails.	
DocuMind AI <i>Hybrid RAG Framework</i>	Jun 2026 – Jul 2026
- Built a hybrid RAG framework using Qwen2.5:7B, LangChain, Ollama, ChromaDB, Pinecone, and vector embeddings. - Developed multi-modal retrieval pipelines supporting PDF, DOCX, PPTX parsing, semantic search, and grounded generation. - Architected a dual-mode inference system enabling local execution, cloud deployment, runtime switching, and response streaming.	
Lead Score Model <i>Mathematical Modeling and Decision Intelligence</i>	May 2026 – Jun 2026
- Built a mathematical lead qualification framework spanning over 10 scoring parameters and 125+ CRM business variables. - Applied Bayesian inference, Beta confidence intervals, nonlinear optimization, and probabilistic modeling for robust lead ranking. - Designed explainable lead prioritization using sigmoid transforms, exponential decay, adaptive score calibration, and optimization.	
Independent Study <i>Riemann Surfaces and Algebraic Curves</i>	May 2026 – Jun 2026
- Studied Riemann surfaces, complex manifolds, and algebraic curves, establishing rigorous foundations in complex geometry. - Analyzed holomorphic mappings, complex atlases, topological genus, and affine plane curves through rigorous formal proofs. - Connected Riemann surface theory with algebraic geometry using meromorphic forms, function fields, and conformal maps.	

Technical Skills

Languages & Frameworks: Python, C, C++, Java, SQL, JavaScript, FastAPI, Streamlit, LangChain, Ollama, LLM Orchestration
AI Infrastructure: LLMs, RAG, Prompt Engineering, Semantic Retrieval, Vector Databases, ChromaDB, Pinecone, Vector Search
Libraries & Tools: NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Git, GitHub, Jupyter Notebook, Tableau, OOP, SQLite, $\text{\LaTeX}$
Mathematical Concepts: Bayesian Inference, Thompson Sampling, Statistical Inference, Nonlinear Optimization, Semantic Parsing

Leadership & Activities

- Won **silver** in **Powerlifting** at the General Sports Championship, IITB and **bronze** at the Overall Institute Powerlifting Opens.
- Coordinated **15+ T20 matches** during Aahvan and served as Hostel **Secretary**, representing over **1200+ hostel residents**.
- Completed Summer of Science in **Data Structures & Algorithms** and Season of Code in **Competitive Programming**.
- Served as **Services** Coordinator at Mood Indigo, **Operations** Coordinator at E-Cell, and **Infrastructure** Coordinator at Techfest.